

P2P Management Insights & Recommendations

Purchase-to-Pay Process Mining & Exception Intelligence

251,656

Analysis Cases

64.04

Median Cycle Days

126.08

P90 Cycle Days

12,361

High-Priority Cases

Management question

Where are P2P transactions experiencing rework, reversals, unusually long cycle times, or process complexity, and which cases or process areas should be investigated first?

Executive summary

The analysis covers **251,656** cases after excluding **78 timestamp-anomaly cases** from duration comparisons. The median case duration is **64.04 days**, while the 90th percentile reaches **126.08 days**, indicating a material long-tail of slower cases. PO modifications occur in **31,656 cases (12.58%)**; invoice reversals in **6,448 (2.56%)**; GR reversals in **2,470 (0.98%)**; and payment-block handling in **55,930 (22.22%)**. A transparent prioritization framework identifies **12,361 cases (4.91%)** for first-line operational review.

The most useful management conclusion is not that every flagged case is defective. The evidence shows where process complexity and exception signals are concentrated so that process owners can focus investigation on a manageable subset of transactions and workflows.

Key decision principles

Signal	Use in decision-making
Long cycle ($\geq$ P90)	Prioritize duration outliers for process review; not an SLA breach by itself.
PO modifications / reversals	Investigate recurring correction patterns and their operational context.
High resource handoffs	Review coordination complexity where longer cycle time is also observed.
High investigation score	Review first; do not interpret as fraud, non-compliance, or supplier fault.

1. Process performance and exception profile

Metric	Cases	Rate	Interpretation
PO modification	31,656	12.58%	Explicit PO change activity present.
Invoice reversal	6,448	2.56%	Case contains Cancel Invoice Receipt.
GR reversal	2,470	0.98%	Case contains Cancel Goods Receipt.
Payment-block handling	55,930	22.22%	Set and/or Remove Payment Block observed.
Repeated activity	22,413	8.91%	One or more activities repeat within the case.

Finding 1 - Long cycle-time tail

The median cycle time is 64.04 days, while P90 is 126.08 days. This means the slowest 10% of valid-timestamp cases take at least about 126 days. A percentile threshold is preferable to an arbitrary 30/60/90-day rule because no contractual SLA is available in the dataset.

Finding 2 - PO modification is associated with more complex processing

Cases containing explicit PO modification events show a higher observed median cycle time than cases without such modifications (approximately 83.0 days versus 62.4 days in the process-analysis notebook). This is an association, not proof that the change caused the delay. The operational question is whether particular modification types, vendors, or process categories repeatedly require avoidable correction.

Finding 3 - Payment-block data requires careful interpretation

Remove Payment Block is far more common than Set Payment Block in the event log. The project therefore reports Set, Remove, and Any Payment Block Handling separately rather than presenting a simplistic block-rate lifecycle. The correct management use is to investigate block-handling cases and time to clearing, not to infer that all blocks originated inside the visible event sequence.

What management should investigate next

- Break PO modification cases down by change type (quantity, price, approval, delivery indicator, storage location, currency, payment term).
- Review long-cycle cases by process category and high-volume vendors rather than applying one threshold to all process types.
- Use case timelines to validate whether repeated activities represent operational rework, legitimate partial processing, or expected workflow behavior.

2. Where process behavior differs

Process category	Cases	Median cycle	Exception rate	PO mod rate
3-way match, invoice before GR	220,949	66.35 d	38.97%	12.31%
3-way match, invoice after GR	15,165	63.39 d	54.24%	13.14%
Consignment	14,498	19.93 d	19.85%	9.84%
2-way match	1,044	23.69 d	100.00%	100.00%

Interpretation: The four item categories are structurally different processes. The 100% PO-modification rate for 2-way match is a strong reason *not* to label every modification as a defect. That pattern should be validated as part of the process design before any corrective action is proposed.

Document-type signal

Document type	Cases	Median cycle	Exception rate	PO mod rate	Repeated activity
Standard PO	248,682	64.05 d	38.79%	12.19%	8.60%
Framework order	1,536	34.56 d	91.28%	79.49%	48.44%
EC Purchase order	1,438	64.47 d	26.70%	7.79%	19.68%

Management implication

Framework orders display substantially different behavior from Standard POs: high exception-signal prevalence, high PO-modification prevalence, and much more repeated activity. This should be treated as a distinct workflow for process-owner validation rather than benchmarked directly against Standard POs.

3. Complexity and investigation prioritization

Resource handoffs	Cases	Median cycle days	Median events
0	9,277	0.05	2
1-2	28,039	17.25	2
3-5	184,987	69.09	5
6-10	27,060	85.72	8
11+	2,293	92.52	41

Observed cycle time rises across handoff bands, and cases with 11+ handoffs also have a much larger event count. This is useful evidence of process complexity, but it does not establish that handoffs themselves caused the delay. High-handoff cases should be reviewed for unnecessary routing, approval loops, or handoff duplication.

Investigation-priority distribution

Priority	Cases	Share	Recommended use
Low	197,765	78.59%	Routine monitoring
Medium	41,530	16.50%	Segment-level review / sampling
High	12,361	4.91%	First-line case investigation

Why the score is defensible

The priority score is intentionally transparent: long cycle and reversals receive 2 points; PO modification, PO intervention, payment-block handling, repeated activity, and high handoffs receive 1 point each. It is a review-order mechanism, not a predictive risk model.

4. Recommended management actions

Priority	Action	Why now	Measure to monitor
1	Review the 12,361 high-priority cases using case timelines and process-owner validation.	Focuses first-line review on 4.91% of cases with multiple transparent signals.	High-priority count; review status; recurring signal mix.
2	Separate process governance by Item Category and Document Type.	2-way match, framework orders, 3-way matching and consignment behave differently.	Cycle time and exception rates within each process type.
3	Investigate recurring PO modification patterns by change type and high-volume vendor.	PO modification occurs in 12.58% of valid-timestamp cases and is associated with longer observed duration.	PO modification rate; median cycle by change type.
4	Review high-handoff cases for avoidable routing or approval loops.	Observed median duration rises across handoff bands.	Median handoffs; high-handoff count; cycle by handoff band.
5	Treat payment-block handling as a separate AP investigation workflow.	Set/Remove asymmetry makes one simple block-lifecycle KPI unreliable.	Set cases; Remove cases; Remove-to-Clear duration.
6	Create a monthly process-review cadence using governed KPI definitions.	Keeps SQL, Python, Excel and Power BI aligned over time.	P90 cycle; exception rates; high-priority backlog.

What this project does not claim

- No flagged case is automatically classified as fraud, non-compliance, supplier failure, or employee underperformance.
- Observed vendor or process differences are descriptive and require denominator-aware comparison and operational validation.
- No ROI, cost savings, working-capital benefit, or financial loss is claimed because the dataset does not provide sufficient evidence for those conclusions.
- Associations between modifications, handoffs, complexity and cycle time are not presented as causal effects.

Portfolio positioning

This project demonstrates an end-to-end analyst workflow: business framing -> event-log validation -> case-level modeling -> SQL/Python analysis -> operational Excel review -> Power BI decision support -> evidence-based recommendations. The strongest interview story is the analytical discipline used to avoid double counting, unsupported benchmarks, and causal overstatement.

Source note: BPI Challenge 2019 Purchase-to-Pay event log converted to CSV. All figures in this report are calculated from the project analytical model. The dataset is anonymized.